

Part (b) Calculate Z-Score for first four observations

Formula:
$$Z = \frac{x - \bar{x}}{\sigma}$$

First, calculate population Standard Deviation

Step 1: Find Squared deviations

Age	$x - 33$	$(x - 33)^2$
18	-15	225
22	-11	121
25	-8	64
42	9	81
28	-5	25
43	10	100
33	2	4
35	23	529
56	-5	25
28		

$$\sum (x - \bar{x})^2 = 1174$$

$$\Rightarrow \text{Standard deviation} = \sigma = \sqrt{\frac{1174}{10}} = \sqrt{117.4} \approx 10.835$$

$$\Rightarrow \text{Z-Score of age 18} = Z = \frac{18 - 33}{10.835} = \frac{-15}{10.835} = -1.384$$

$$\Rightarrow \text{Z-Score of age 22} = Z = \frac{22 - 33}{10.835} = \frac{-11}{10.835} = -1.015$$

$$\Rightarrow \text{Z-Score of age 25} = Z = \frac{25 - 33}{10.835} = \frac{-8}{10.835} = -0.738$$

$$\Rightarrow \text{Z-Score of age 42} = Z = \frac{42 - 33}{10.835} = \frac{9}{10.835} = 0.831$$

⑤ Standardization of Patient Age data:

Given ages: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Part (a): Mean Absolute Deviation (MAD)

Step 1: find Mean

Formula $\bar{x} = \frac{\sum x}{n}$

$$\bar{x} = \frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28}{10}$$

$$= \frac{330}{10}$$

$$\bar{x} = 33$$

Step 2 $\Rightarrow$ find Absolute deviations

Age x	Mean $\bar{x}$	$ x - \bar{x} $	$ x - \bar{x} $
18	33	15	15
22	33	11	11
25	33	8	8
42	33	9	9
28	33	5	5
43	33	10	10
33	33	0	0
35	33	2	2
56	33	23	23
28	33	5	5

$$15 + 11 + 8 + 9 + 5 + 10 + 10 + 0 + 2 + 23 + 5 = 88$$

Step 3 $\Rightarrow$ Calculate MAD

$$MAD = \frac{\sum |x - \bar{x}|}{n} = \frac{88}{10} = 8.8$$

Step 4: Create Bins

Bin 1: 18.5 to 25.5

Values: 18.5, 19.2, 21.8, 24.9, 25.5

Boundaries: 18.5, 25.5

Bin 2: 25.5 to 32.5

Values: 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4

Boundaries: 25.5, 32.5

Bin 3: 32.5 to 39.5

Values: 33.1, 33.6, 34.7, 38.2, 39.5

Boundaries: 32.5, 39.5

4. DATA REDUCTION USING BIN MEDIANS

Given Data:

120, 150, 160, 160, 140, 145, 148, 150, 150, 155, 155,
155, 155, 155, 165, 170, 170, 175, 175, 175, 186, 190,
200, 210, 230, 300

Step 1: Sort the data

$$\therefore n = 28$$

Assume 4 Bins

$$\frac{28}{4} = 7$$

$\therefore$ Therefore each Bin contains 7 values.

Step 2: Create Bins

Bin 1 : 120, 140, 145, 145, 148, 150, 150

Middle Value = 4th Value

Median = 145

Replace all values by 145:

Bin 2 : 150, 155, 155, 155, 155, 155, 160

Middle Value = 4th Value

Median = 155

Replace all values by 155:

Bin 3: $\Rightarrow$ 160, 165, 170, 170, 175, 175, 175

Middle Values = 4th Value

Median = 170

Replace all by 170:

Bin 4: $\Rightarrow$ 175, 175, 180, 190, 200, 210, 230

Median = 190

Replace all values by 190:

The Value 300 is Left Outside Because the Original data Contains 28 Values and four equal frequency Bins of 7 Values already account for all 28 values therefore the Correct Bin is 4.

175, 180, 190, 200, 210, 230, 300

$\Rightarrow$ Median = 200

Step 4: Replace values with Bin means:

Bin 1 $\Rightarrow$ 115.71 (7 times)

Bin 2 $\Rightarrow$ 124.86 (7 times)

Bin 3 $\Rightarrow$ 137 (6 times)

Bin 4 $\Rightarrow$ 154.5 (6 times)

3 Equal - Width Binning For Risk Categorization

Given Sorted BMI data:

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2,
30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5.

Step 1: Find Minimum and Maximum

Minimum = 18.5

Maximum = 39.5

Step 2: Calculate Range

$$\begin{aligned} \text{Range} &= 39.5 - 18.5 \\ &= 21 \end{aligned}$$

Step 3: Calculate Bin Width

Formula:
$$\text{Bin Width} = \frac{\text{Range}}{\text{Number of Bins}} = \frac{21}{3} = 7$$

$$\text{Bin Width} = 7$$

Step 1: Count Observation

$$n = 26$$

Assume we use 4 Bins

$$n = 26$$

$$\frac{26}{4} = 6.5$$

Step 2: Create Equal-frequency Bins

$$\text{Bin 1} \Rightarrow 110, 112, 115, 115, 118, 120, 120 \Rightarrow 7$$

$$\text{Bin 2} \Rightarrow 122, 124, 124, 126, 126, 126, 130 \Rightarrow 7$$

$$\text{Bin 3} \Rightarrow 135, 135, 138, 138, 138, 138 \Rightarrow 6$$

$$\text{Bin 4} \Rightarrow 140, 145, 150, 152, 160, 180 \Rightarrow 6$$

Step 3: Find the mean of each Bin

$$\text{Bin 1} \Rightarrow \text{Mean} = \frac{110 + 112 + 115 + 115 + 118 + 120 + 120}{7} = \frac{810}{7}$$

$$\text{Mean} = 115.71$$

$$\text{Bin 2} \Rightarrow \text{Mean} = \frac{122 + 124 + 124 + 126 + 126 + 126 + 130}{7} = \frac{874}{7}$$

$$\text{Mean} = 124.86$$

$$\text{Bin 3} \Rightarrow \text{Mean} = \frac{135 + 135 + 138 + 138 + 138 + 138}{6} = \frac{822}{6}$$

$$\text{Mean} = 137$$

$$\text{Bin 4} \Rightarrow \text{Mean} = \frac{140 + 145 + 150 + 152 + 160 + 180}{6} = \frac{927}{6}$$

$$\text{Mean} = 154.5$$

$$\begin{aligned}\sum (x - \bar{x})(y - \bar{y}) &= 5.2875 + 0.5625 + 6.1875 + 0.2125 \\ &= 12.25\end{aligned}$$

Step 4: Calculate Covariance

FORMULA,

$$\text{Cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n}$$

$$= \frac{12.25}{4} = 3.0625$$

$$\text{Cov}(x, y) = 3.0625$$

$\therefore$ Since Covariance is Positive there is a Positive Relationship Between Premium Increase and claim frequency.

CONCLUSION:

When the premium increase rate increase the number of claims also tends to increase

2. DATA Smoothing Using Equal Frequency Binning

Given DATA:

110, 112, 115, 115, 118, 120, 120, 122, 124, 126, 126, 126,
130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

① Covariance Analysis In Health Insurance

x_i	y_i
3.2	15
4.0	18
5.5	25
4.8	20

We need to find?

1. Mean of x
2. Mean of y
3. Covariance
4. Whether Relationship is + or -.

Step 1: Find the Mean of x :

$$\bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4} = \frac{17.5}{4}$$

$$\bar{x} = 4.375 \Rightarrow \text{Mean}$$

Step 2: Find the mean of y :

$$\bar{y} = \frac{15 + 18 + 25 + 20}{4} = \frac{78}{4}$$

$$\bar{y} = 19.5 \Rightarrow \text{Mean}$$

Step 3: Calculate Standard Deviation

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125